NHS

Capacity & Resource Utilisation Report

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Introduction

The National Health Service faces the critical task of effectively managing its resources amid growing demand. Our team conducted comprehensive exploratory data analysis using provided internal data, focusing on resource utilisation, staff adequacy, missed GP appointments and trends over time. This report summarises our key findings, identifies prominent trends and provides actionable recommendations for the NHS.

Data Overview

Three datasets were provided for exploratory analysis, each beginning at different periods but all ending in June 2022. One of the datasets stretched back to the beginning of 2020, allowing us to analyse the change in total appointments and method of delivery as the COVID-19 pandemic emerged. The three datasets comprised appointments from 42 distinct Integrated Care Board (ICB) ONS codes, further segmented in 106 sub-locations, encompassing a variety of operational contexts and patient demographics. There were 5 unique service settings within the data, while appointment statuses were categorised into attended, did not attend (DNA) and unknown. A fourth data extract was provided from Twitter (X), however, due to the challenges in analysis, it yielded limited insights.

Appointment Distribution

Our analysis clearly indicated that the General Practice setting manages the most appointments (92%)[1], while 83% were labelled as a care related encounter. We found that the vast majority of appointments took place between Monday to Friday[2], although there were several outliers. During the analysed period, consistent seasonal fluctuations emerged, revealing peak demands often associated with winter months due to increased cases of flu and other respiratory illnesses[3]. Overall, there is a positive trend of appointments over the analysed period, however, we would recommend looking at a longer period confirm this result.

Regional analysis of appointments highlighted differences in resource demand. Of the six regions captured in the analysis, the Midlands recorded the highest overall appointment volume, followed by the North-East and Yorkshire. The South-West of England recorded the fewest appointments over the period. At the sub-location level, NE/N Cumbria recorded the most appointments by a clear margin[4]. We note, however, there were inconsistencies between data sources as a result of different periods in question.

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[1] Figure 1 – Total appointments by NHS Service setting
[2] Figure 2 – Distribution of appointments between Monday - Sunday
[3] Figure 3 – Distribution of appointment by month
[4] Figure 4 – Total Appointments by region and location

The period under analysis was notably affected by the COVID-19 pandemic, influencing appointment scheduling, attendance patterns and service delivery methodologies. The introduction of lockdown measures led to temporary decreases in appointments, particularly visible in early 2020. Moreover, we found that there was a surge in telephone appointments between March 2020 and June 2020, coinciding with the first lockdown period in the UK[5], demonstrating the adaptability of services provided by GPs.

Capacity

Appointment duration evaluation suggests NHS resources have generally been utilised effectively, particularly in managing short-duration appointments. A significant proportion of appointments lasting typically 1-10minutes highlighted efficient practices[6]. However, 24% of appointments were categorised under "Unknown/Data Quality" indicating a critical data collection gap. Additionally, this data has only been captured since April 2022 and is recorded differently depending on the practice's system supplier.

The NHS forecast an average of 1,200,000 appointments per day as a guideline for maximum capacity. We have found that out of 334 days from 1st August 2021 to 30th June 2022, there were 175 days where this maximum was breached[7]. These breaches place additional strain on staffing and resources.

Missed Appointments

Missed appointments remained a persistent issue, representing approximately 3-6% of total appointments across sub-locations throughout the analysed period[8]. We found that despite being the 18th highest sub location by total appointments, the Black Country had the highest location for appointments that were not attended. The significant variation in DNA rates observed between different geographic regions implies the influence of factors such as healthcare accessibility and transportation challenges.

Interestingly, during periods of reduced total appointment volumes, particularly March 2020 to August 2020, a concurrent reduction in DNA percentages was observed[9]. This suggests that improved attendance may have been influenced by a greater proportion of urgent appointments during these periods, increased accessibility and convenience through remote consultations.

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[5] Figure 5 – Percentage of appointments via Telephone/Online by month
[6] Figure 6 – Appointments by duration
[7] Figure 7 – Frequency of daily appointments.
[8] Figure 8 – DNA appointments by ICB Location
[9] Figure 9 – DNA appointments by month

There was also a correlation between missed appointments and healthcare professionals[10]. Of the appointments that were missed, those provided by '*Other healthcare professionals'* were higher than GPs. However, we do note that there were limitations to this data collection.

Scenario analysis

In undertaking our analysis, we calculated the total number of appointments that occurred each day, once missed appointments were taken into account[11]. For this, we analysed two estimates of DNA percentages, 3% and 6%. We found this made a significant impact on the frequency of appointments per day. After adjusting for 6% of missed appointments, the average frequency of appointments per day fell below the maximum capacity, while 3% was broadly in line with the maximum capacity threshold.

Conclusion

This analysis highlights several key areas of concern and opportunity for the NHS in relation to capacity management and resource utilisation. While the system has demonstrated significant adaptability, the persistent occurrence of missed appointments challenges operational efficiency.

The breaching of forecasted capacity on over half the days reviewed signals mounting strain on healthcare resources. However, our scenario analysis shows that if missed appointments were effectively reduced, daily volumes would likely approach or exceed operational thresholds more consistently, creating a need for proactive capacity planning. This underscores a critical trade-off: while reducing DNA rates is essential for optimising patient care and minimising resource waste, it must be done in parallel with a strategic review of workforce and infrastructure planning to accommodate the additional load.

Furthermore, regional disparities in attendance patterns and service demand point to the need for tailored, location-specific strategies. Addressing factors such as transport accessibility, digital inclusion, and targeted community engagement could support more equitable service delivery across Integrated Care Boards. Analysing the correlation between population demographics and total appointments by sub-location could provide insights into the factors influencing appointments and locations to target capacity issues.

Lastly, improvements in data consistency are vital. Reliable data underpins effective decision-making. As such, we recommend investment in standardised data collection practices across systems to support future forecasting, performance monitoring, and service delivery optimisation.

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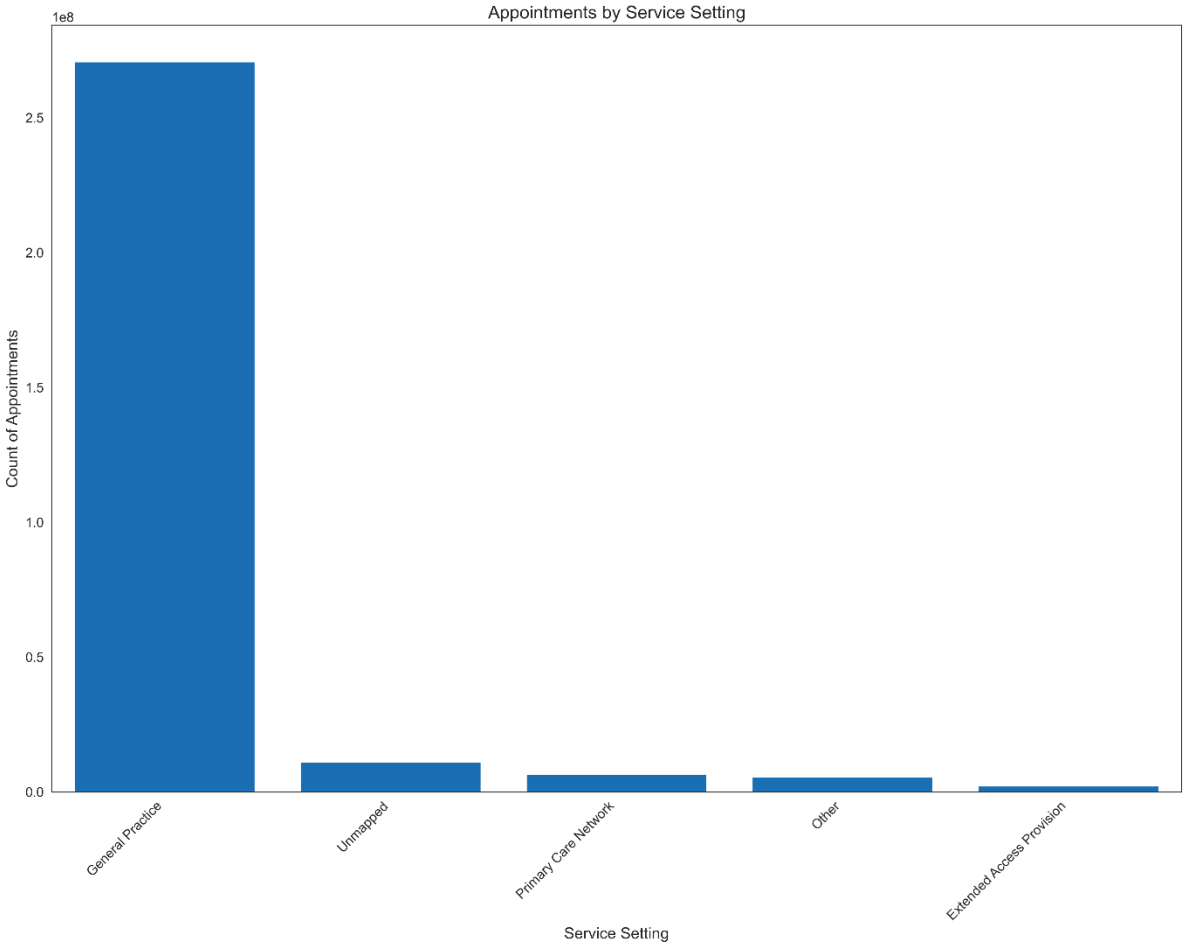
[10] Figure 10 – Total appointments by status and healthcare professional

[11] Figure 10 – Adjusted frequency of daily appointments.

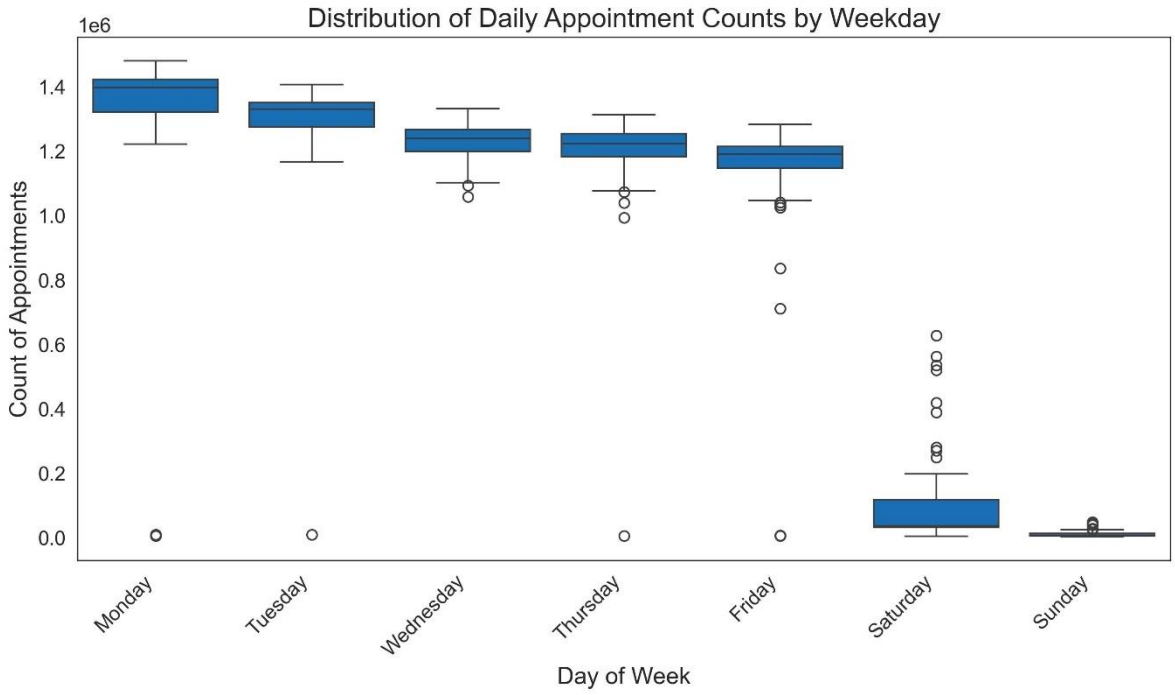
Appendix

Figure 1 – Appointments by service setting



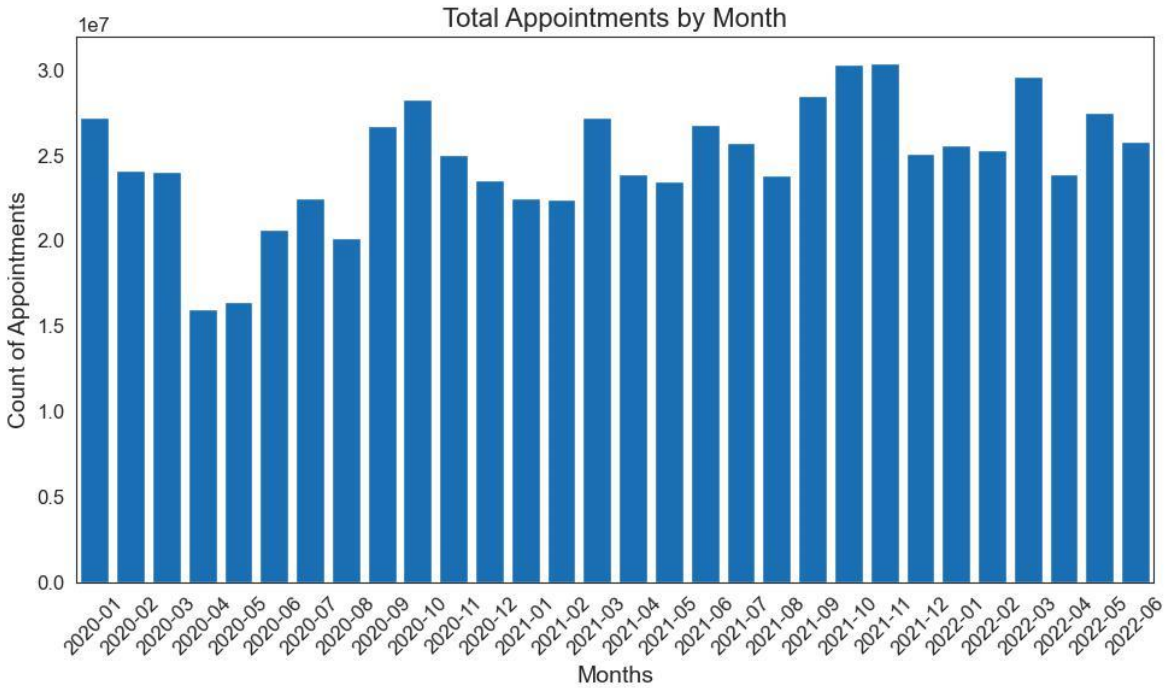
Source: national_categoies.xlsx

Figure 2 – Distribution of appointments over the week



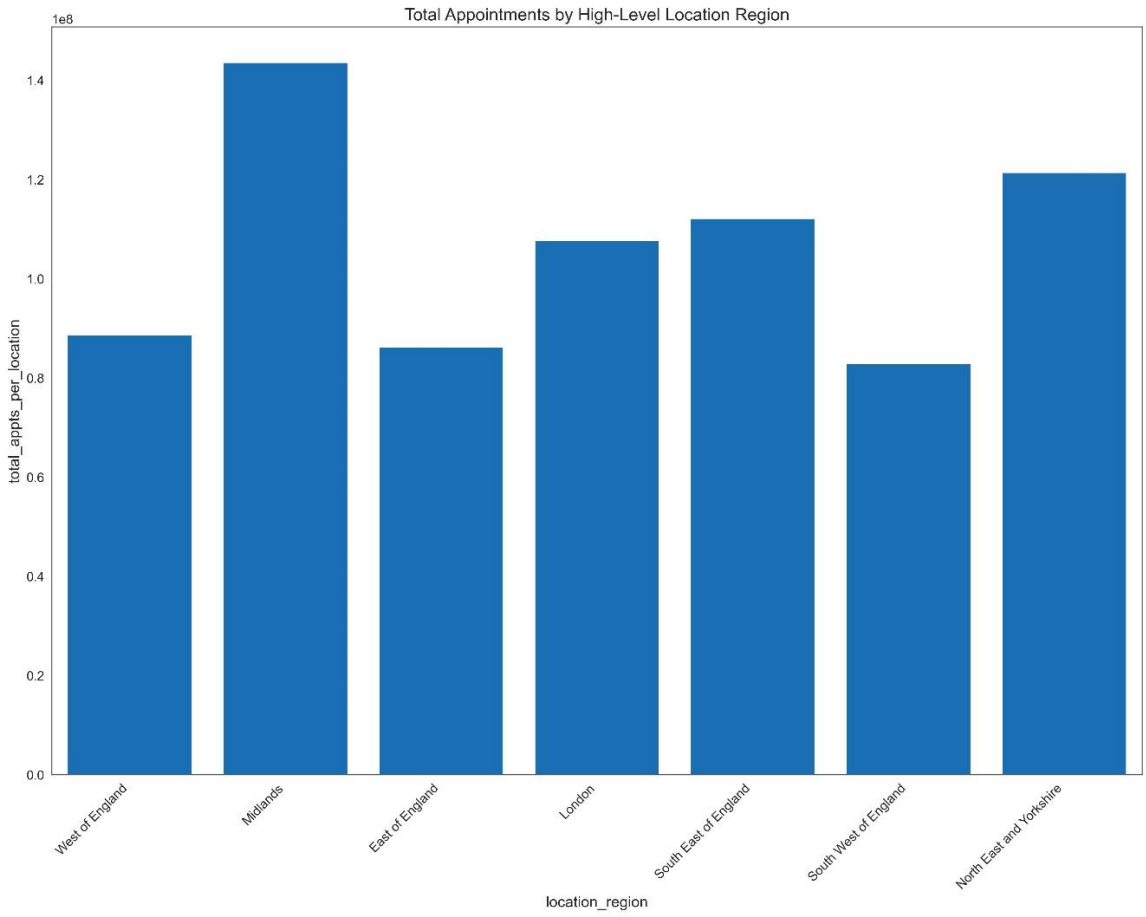
Source: actual_duration.csv

Figure 3 - Disitribution of Appointments by Month

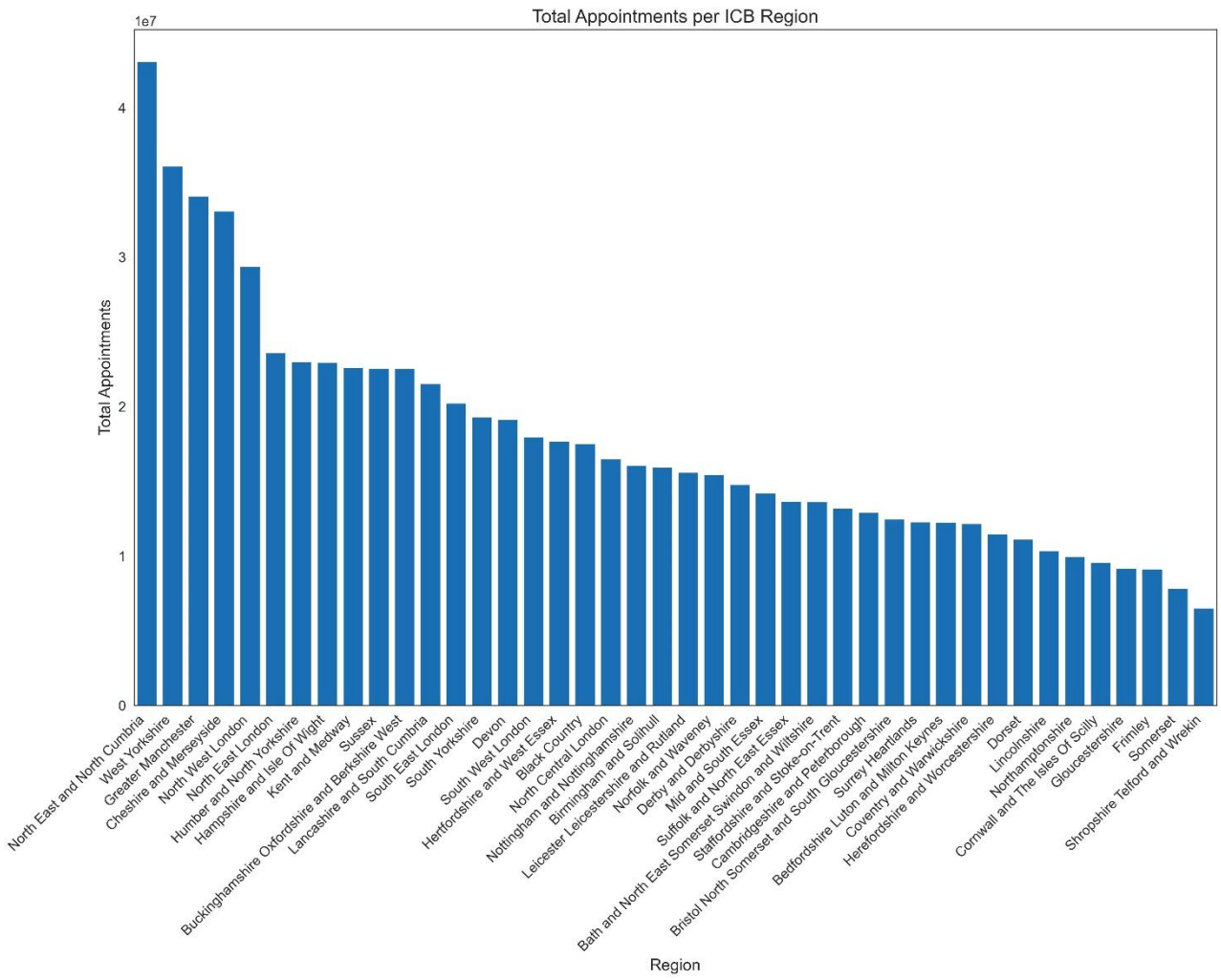


Source: appointments_regional.csv

Figure 4 – Total Appointments by region and location

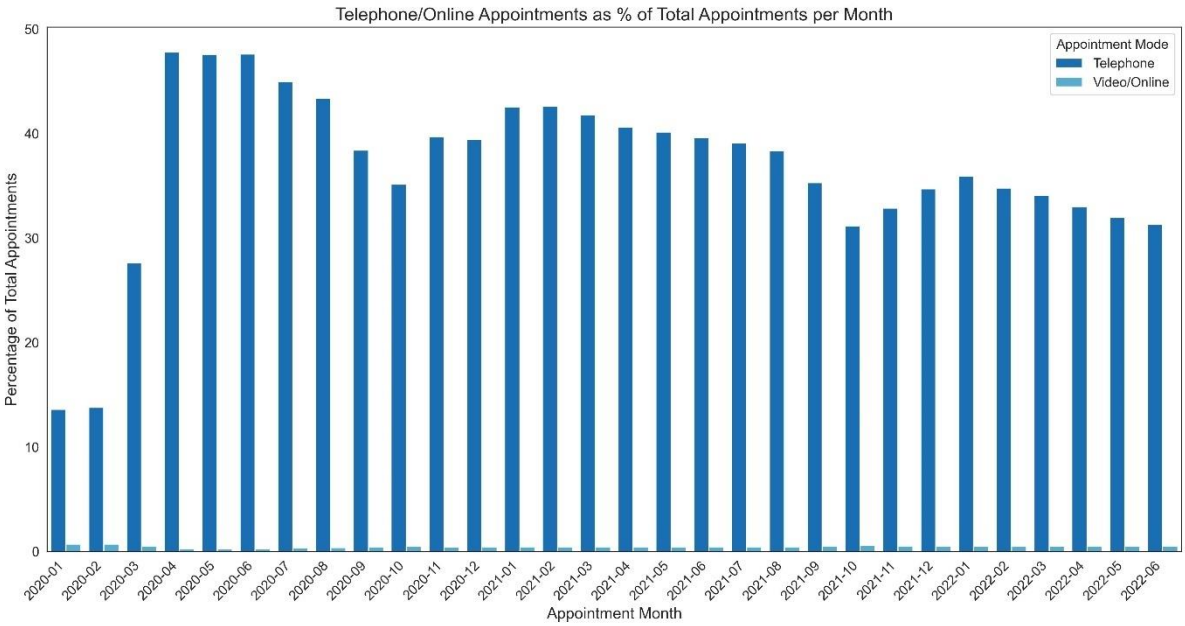


Source: appointments_regional.csv



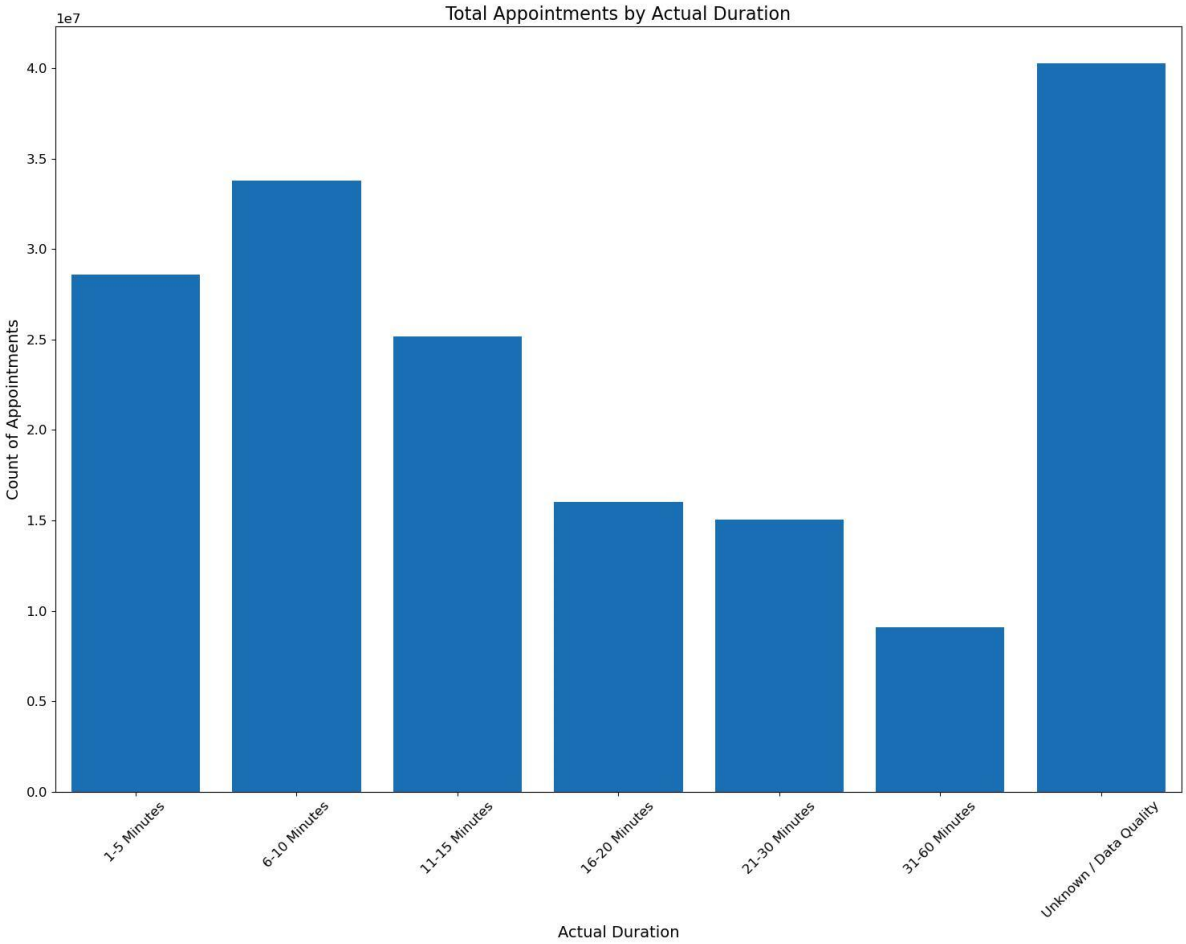
Source: appointments_regional.csv

Figure 5 – Monthly appointments by Telephone



Source: appointments_regional.csv

Figure 6 – Appointments by duration



Source: actual_duration.csv

Figure 7 – Frequency of daily appointments

Please note, this chart includes appointments from Monday – Friday and outliers are filtered by interquartile range.

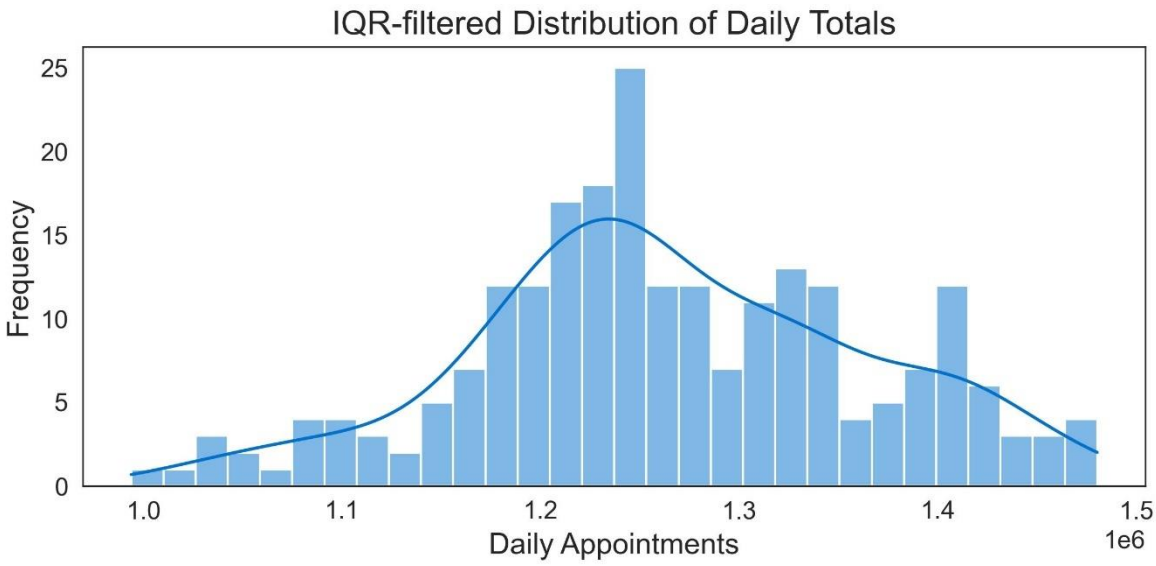
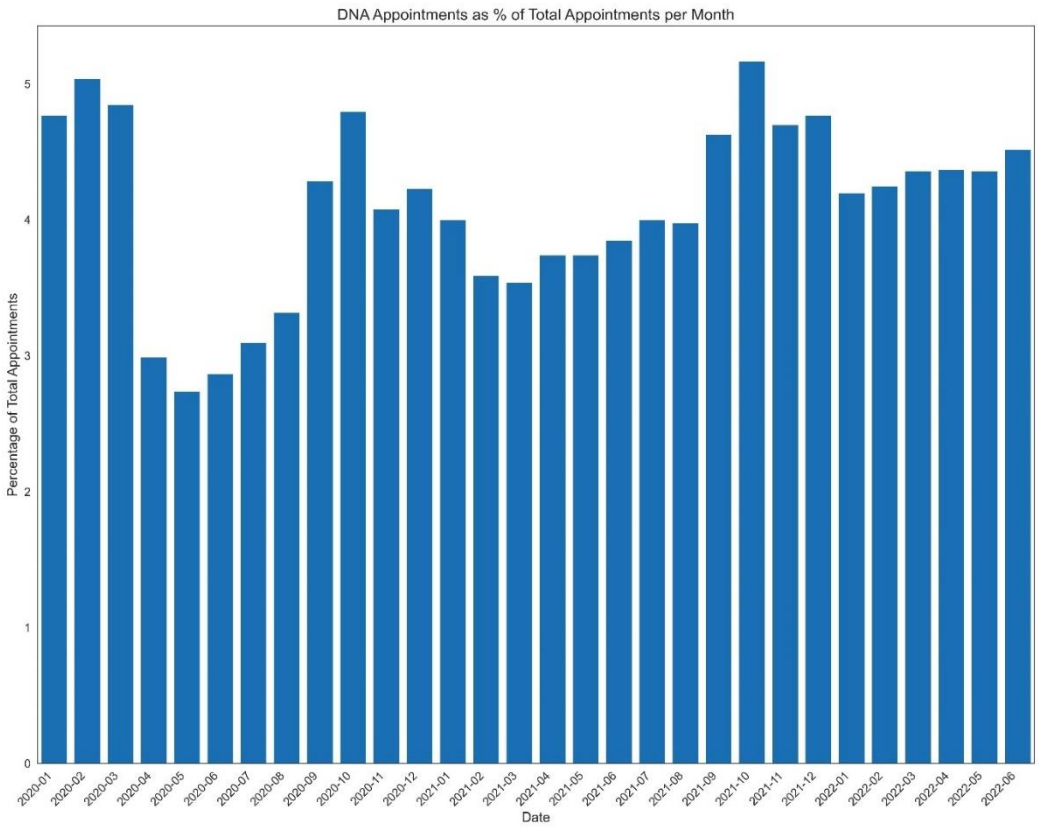


Figure 8 – DNA appointments per month



Source: appointments_regional.csv

Figure 9 – DNA appointments by sub-location

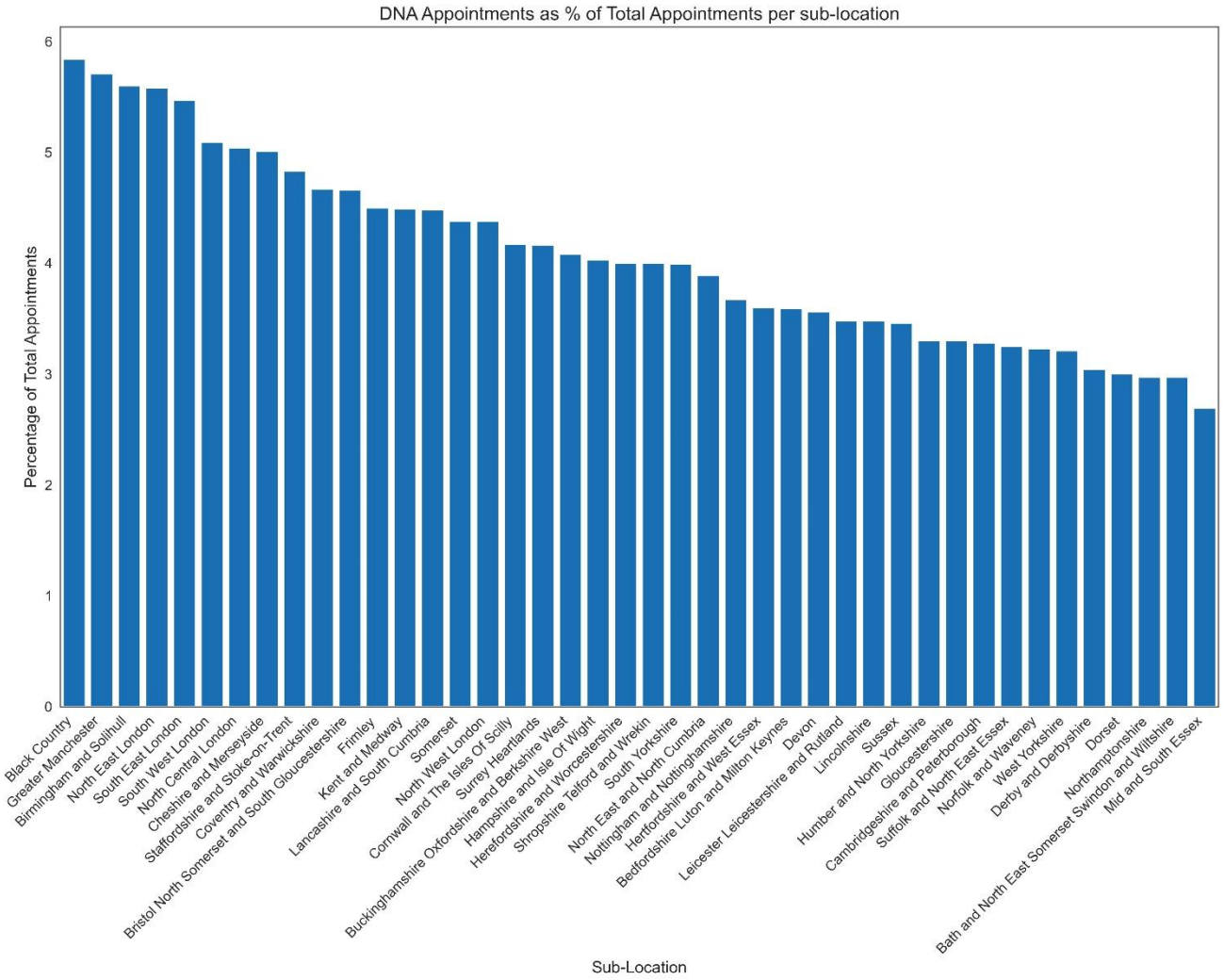
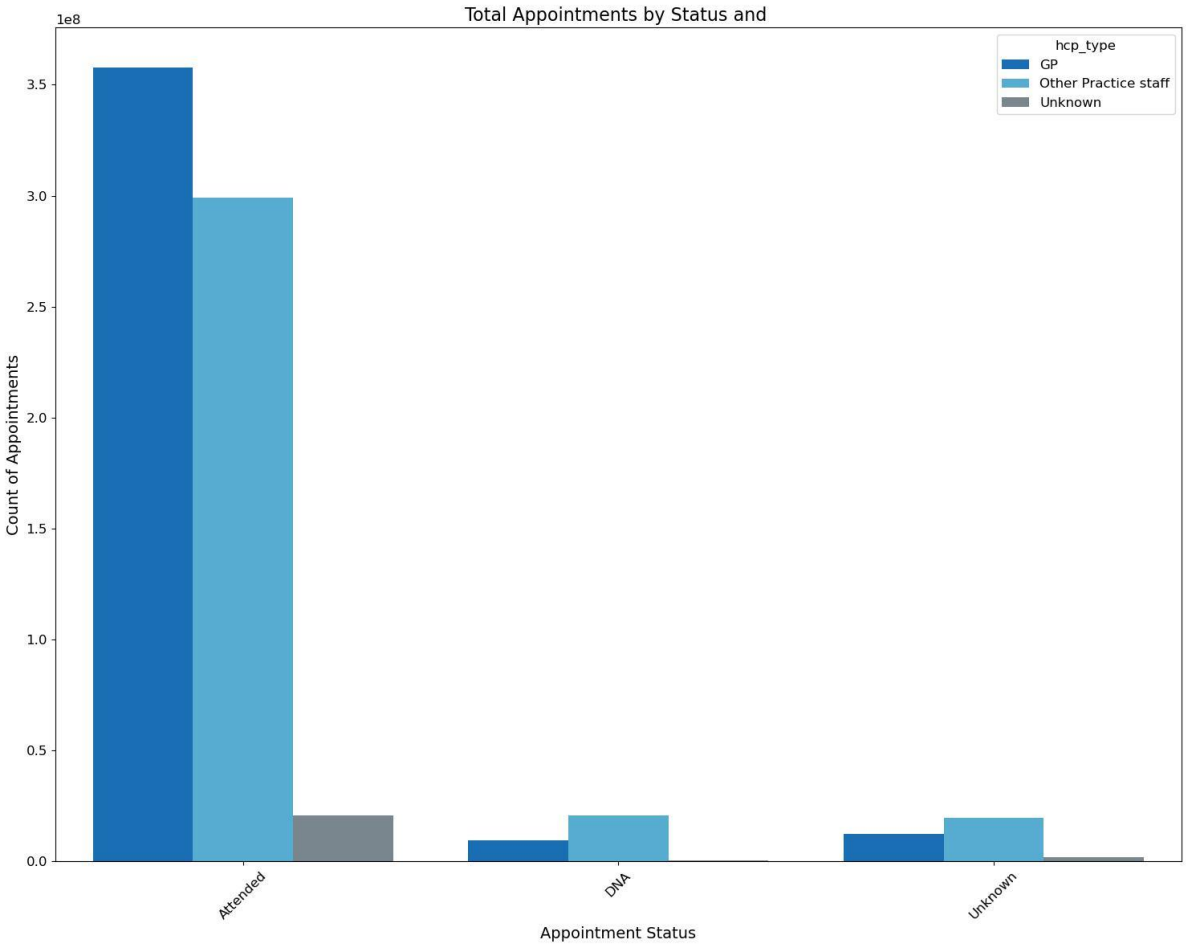
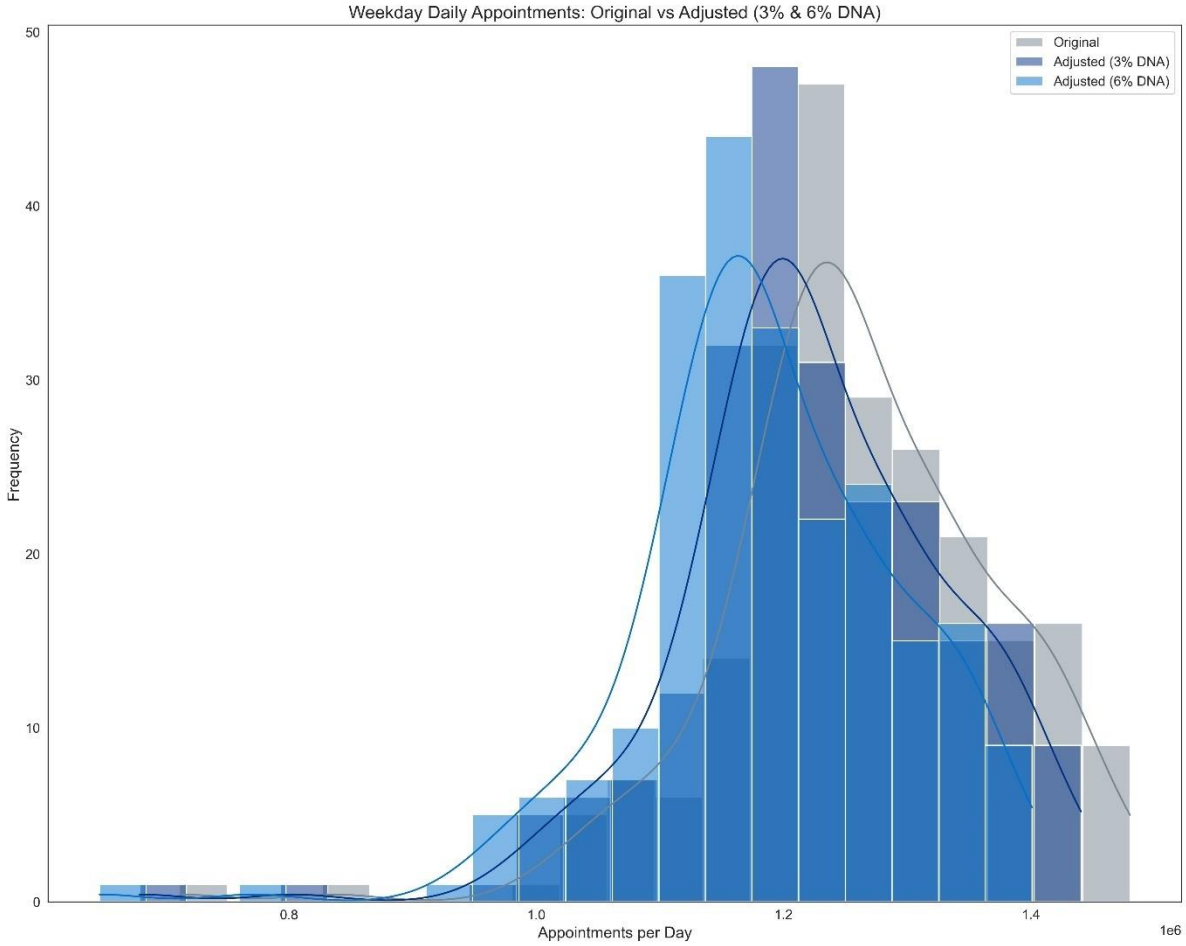


Figure 10 – Appointment by Status and healthcare professional



Source: appointments_regional.csv

Figure – 11 Adjusted frequency of daily total appointments



Source: national_categories.xlsx